

MedGemma recall benchmark: MDACE billing extraction

How much of the billed evidence a 4B open model recovers from a clinical note.

EXECUTION PLAN
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WHAT IS BEING MEASURED

MedGemma reads a note and extracts medical findings. **It does not produce codes.** Each extracted term then goes to the SNOMED lookup, which returns the adjacent ICD code. Asking a model for the code directly compounds two hard problems into one and produces too many errors.

Extraction is therefore the ceiling on the whole pipeline: the lookup can never recover a code whose term was never extracted. This benchmark measures where that ceiling sits for **MedGemma-4B, zero-shot**. There is no target to hit. It reports the best the model achieves, and Ehtesham Bhai decides whether that clears the bar.

DATA

8-07-mdace-ner-eval_sample_100-LOCAL.jsonl, and nothing else. Every note in the file is used.

24	100	91
NOTES	ANNOTATION ROWS	DISTINCT CODES

WHAT COUNTS AS A CORRECT EXTRACTION

Three sources describe each billed code. **Matching any one of them counts**, and each is also scored on its own.

Evidence text	what the note says (<i>depression</i>)
Code description	ICD catalogue wording (<i>Major depressive disorder, single episode, unspecified</i>)
SNOMED terms	clinical synonyms for the same concept

That is a median of **4 accepted phrasings per code**. Recall is reported both per row (100) and per distinct code (91).

HOW A MATCH IS DECIDED

Matching is approximate, not exact. Each level contains the one above it, and **every figure is reported at every level**, so the cost of each degree of looseness is visible rather than hidden inside one threshold.

LEVEL	RULE	CATCHES
L1 exact	identical after normalising	baseline
L2 contains	one is a whole-word subset of the other	back pain in chronic back pain
L3 similar	word overlap and character similarity	typos, word order
L4 embeddings	biomedical sentence model, cosine	HTN to hypertension
L5 LLM judge	a model decides whether two phrases mean the same	adjudication

L4 is the level that earns its keep. Abbreviations are the largest single source of missed matches, and no amount of string similarity reaches them: *CHF* and *congestive heart failure* share almost no characters. Only semantic matching closes that gap.

The model is also asked for two things per finding: the phrase as written in the note, and the standard clinical name for it. Either may match, and the standard name is what the lookup would receive in production anyway.

HOW IT RUNS

- Extract once.** MedGemma reads all 24 notes and its output is saved.
- Score five times.** Every matching level runs against that same saved output, so the ladder costs no extra model time.
- Set the thresholds from the data.** L3 and L4 cut-offs are chosen by inspecting the pairs each level newly accepts, not picked in advance.
- Adjudicate with L5** where a level's judgement is contested, and report which level the headline figure came from.

WHAT IS REPORTED

Recall	of the billed codes, how many were found
False positives	count and rate: terms produced that matched nothing
Terms per note	the volume the recall was bought with
Not-in-note rate	hallucination: terms that are not in the text at all

Broken out by source, so it is clear which of the three the model's wording actually lands on:

SCORED AGAINST	RECALL	FALSE POS.
Evidence text	•	•
Code description	•	•
SNOMED terms	•	•
All three combined	•	•

A term matching only the catalogue wording is a hit against the description row and a false positive against the evidence-text row. Only the combined row's false positives matched nothing at all. SNOMED figures are against the synonyms the file carries.

THE NUMBER THAT MAKES RECALL MEANINGFUL

With loose matching and no volume control, a model that lists every phrase in the note scores near 1.00. Recall alone therefore cannot rank two models: a smaller, more verbose one would appear to beat a larger, more disciplined one.

So recall is always quoted **with the terms-per-note it was bought with**: “0.72 at 20 terms per note”, never a bare 0.72. That is what keeps the benchmark usable when a larger model is measured against it later, and it is why false positives are reported alongside.

WHAT LIMITS THE NUMBER

SNOMED coverage in the file is partial. It carries 24,333 concept terms where the count field reports 270,287 in total, so roughly 9% are present, and procedure codes (CPT and ICD-10-PCS) carry none. Recall against that source is therefore a floor: some synonyms that would have matched are not there to match.

Approximate matching can be wrong in both directions. *Acute* and *chronic renal failure* differ by one word and are clinical opposites; a negated mention reads as a positive one. Recall rises at every level *by construction*, so L4 will always look best. L5 is what decides whether those extra matches are genuine.

STARTING POINT

Scored the strict way, exact matching against the note wording alone, MedGemma returned **recall 0.53** on an initial run, against the ≈0.5 Ehtesham Bhai observed independently. That is the floor this plan is designed to lift, and every change above widens what counts as correct without touching the model itself.

Scope. Extraction only. ICD code prediction is a separate experiment and is not measured here.

MedGemma-4B-it · 4-bit · greedy · zero-shot